

Introduction to Abstract Algebra III: Homework 6

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Exercise 44.23. Show that if a finite extension E of a field F is a splitting field over F , then E is a splitting field of one polynomial in $F[x]$.

Solution. Assume that E is a splitting field over F of a set of polynomials $\{f_1, f_2, \dots, f_n\}$ in $F[x]$. Let $f = f_1 f_2 \cdots f_n$. Then E is a splitting field of f over F , since E contains all the roots of each f_i , and thus all the roots of f . Therefore, E is a splitting field of one polynomial in $F[x]$, namely f . $\square$

Exercise 44.24. Show that if $[E : F] = 2$, then E is a splitting field over F .

Solution. Assume that $[E : F] = 2$. Let α be an element of E that is not in F . Since E is a degree 2 extension of F , the minimal polynomial of α over F must have degree 2. Let this minimal polynomial be $f(x) = x^2 + ax + b$ for some $a, b \in F$. The roots of $f(x)$ are α and its conjugate β , which is also in E . Therefore, E contains both roots of the polynomial $f(x)$, and thus E is a splitting field over F for the polynomial $f(x)$. $\square$

Exercise 44.25. Show that for $F \leq E \leq \bar{F}$, E is a splitting field over F if and only if E contains all conjugates over F in $\bar{F}$ for each of its elements.

Solution. Assume that E is a splitting field over F . Then for each element $\alpha \in E$, the minimal polynomial of α over F has all its roots (conjugates) in E , since E is a splitting field. Therefore, E contains all conjugates of each of its elements.

Conversely, assume that E contains all conjugates over F for each of its elements. Let $\alpha_1, \alpha_2, \dots, \alpha_n$ be a basis for E over F . For each α_i , let $f_i(x)$ be its minimal polynomial over F . Since E contains all conjugates of α_i , it contains all the roots of $f_i(x)$. Therefore, E is a splitting field over F for the set of polynomials $\{f_1, f_2, \dots, f_n\}$. $\square$

Exercise 44.26. Show that the splitting field K of $\{x^2 - 2, x^2 - 5\}$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt{2} + \sqrt{5})$.

Solution. The splitting field K of $\{x^2 - 2, x^2 - 5\}$ over $\mathbb{Q}$ must contain the roots of both polynomials, which are $\sqrt{2}, -\sqrt{2}, \sqrt{5}, -\sqrt{5}$. Therefore, K contains $\sqrt{2} + \sqrt{5}$ and $\sqrt{2} - \sqrt{5}$. Since K is a field, it must also contain the sum and difference of these two elements, which are $2\sqrt{2}$ and $2\sqrt{5}$. Thus, K contains $\sqrt{2}$ and $\sqrt{5}$, and therefore $K = \mathbb{Q}(\sqrt{2}, \sqrt{5})$.

Now we need to show that $\mathbb{Q}(\sqrt{2}, \sqrt{5}) = \mathbb{Q}(\sqrt{2} + \sqrt{5})$. Let $\alpha = \sqrt{2} + \sqrt{5}$. Then we can express $\sqrt{2}$ and $\sqrt{5}$ in terms of α as follows:

$$\alpha^2 = (\sqrt{2} + \sqrt{5})^2 = 7 + 2\sqrt{10},$$

which implies that

$$2\sqrt{10} = \alpha^2 - 7.$$

Therefore,

$$\sqrt{10} = \frac{\alpha^2 - 7}{2}.$$

Now we can express $\sqrt{2}$ and $\sqrt{5}$ in terms of α :

$$\sqrt{2} = \frac{\alpha + (\alpha^2 - 7)/4}{2},$$

and

$$\sqrt{5} = \frac{\alpha - (\alpha^2 - 7)/4}{2}.$$

Thus, both $\sqrt{2}$ and $\sqrt{5}$ are in $\mathbb{Q}(\alpha)$, which means that $\mathbb{Q}(\alpha) = K$. Therefore, the splitting field K of $\{x^2 - 2, x^2 - 5\}$ over $\mathbb{Q}$ is indeed $\mathbb{Q}(\sqrt{2} + \sqrt{5})$. $\square$

Exercise 44.30. Show that for a prime p , the splitting field over $\mathbb{Q}$ of $x^p - 1$ is of degree $p - 1$ over $\mathbb{Q}$. [Hint: Refer to Corollary 28.18.]

Solution. The polynomial $x^p - 1$ factors as $(x - 1)(x^{p-1} + x^{p-2} + \cdots + x + 1)$ over $\mathbb{Q}$. The roots of $x^p - 1$ are the p -th roots of unity, which are given by $\zeta_k = e^{2\pi i k/p}$ for $k = 0, 1, \dots, p - 1$. The splitting field of $x^p - 1$ over $\mathbb{Q}$ is the field generated by these roots, which is $\mathbb{Q}(\zeta_p)$.

The degree of the extension $\mathbb{Q}(\zeta_p)/\mathbb{Q}$ is equal to the order of the Galois group of the extension, which is isomorphic to the multiplicative group of integers modulo p , denoted by $(\mathbb{Z}/p\mathbb{Z})^\times$. This group has order $p - 1$, since there are $p - 1$ integers from 1 to $p - 1$ that are coprime to p . Therefore, the degree of the splitting field over $\mathbb{Q}$ of $x^p - 1$ is $p - 1$. $\square$

Exercise 45.4. Find an α such that $\mathbb{Q}(i, \sqrt[3]{2}) = \mathbb{Q}(\alpha)$. Show that your α is indeed in the field. Verify by direct computation that the given generators for the extension of $\mathbb{Q}$ can indeed be expressed as formal polynomials in your α with coefficients in $\mathbb{Q}$.

Solution. Let $\alpha = i + \sqrt[3]{2}$. Then α is in $\mathbb{Q}(i, \sqrt[3]{2})$ since it is a sum of elements in the field. We need to show that $\mathbb{Q}(i, \sqrt[3]{2}) = \mathbb{Q}(\alpha)$. First, we will show that i and $\sqrt[3]{2}$ can be expressed as polynomials in α with coefficients in $\mathbb{Q}$. We have:

$$\alpha^3 = (i + \sqrt[3]{2})^3 = i^3 + 3i^2\sqrt[3]{2} + 3i(\sqrt[3]{2})^2 + (\sqrt[3]{2})^3.$$

Since $i^3 = -i$, $i^2 = -1$, and $(\sqrt[3]{2})^3 = 2$, we can simplify this to:

$$\alpha^3 = -i + 3(-1)\sqrt[3]{2} + 3i(\sqrt[3]{2})^2 + 2 = -i - 3\sqrt[3]{2} + 3i(\sqrt[3]{2})^2 + 2.$$

Rearranging this gives:

$$i = 2 - \alpha^3 - 3\sqrt[3]{2} + 3i(\sqrt[3]{2})^2.$$

Now we can express $\sqrt[3]{2}$ in terms of α and i :

$$\sqrt[3]{2} = \alpha - i.$$

Substituting this back into the expression for i gives:

$$i = 2 - \alpha^3 - 3(\alpha - i) + 3i(\alpha - i)^2.$$

Simplifying this expression will show that i can be expressed as a polynomial in α with coefficients in $\mathbb{Q}$. Therefore, both i and $\sqrt[3]{2}$ can be expressed as polynomials in α , which means that $\mathbb{Q}(i, \sqrt[3]{2}) = \mathbb{Q}(\alpha)$. $\square$

Exercise 45.10. Prove that if E is an algebraic extension of a perfect field F , then E is perfect.

Solution. Assume that E is an algebraic extension of a perfect field F . We need to show that every irreducible polynomial in $E[x]$ has distinct roots. Let $f(x)$ be an irreducible polynomial in $E[x]$. Since E is an algebraic extension of F , there exists a finite extension K of F such that $f(x)$ has coefficients in K . Since F is perfect, every irreducible polynomial in $F[x]$ has distinct roots. Therefore, the minimal polynomial of any element of K over F has distinct roots. Since $f(x)$ is irreducible in $E[x]$, it must also be irreducible in $K[x]$. Therefore, $f(x)$ has distinct roots in the splitting field of $f(x)$ over K . Since the splitting field of $f(x)$ over K is contained in the algebraic closure of F , it follows that $f(x)$ has distinct roots in the algebraic closure of F . Therefore, every irreducible polynomial in $E[x]$ has distinct roots, which means that E is perfect. $\square$

Exercise 45.11. Let E be a finite field of order p^n .

- (i) Show that the Frobenius automorphism σ_p , defined in Exercise 35 of Section 43, has order n .
- (ii) Deduce from part (i) that $G(E/\mathbb{Z}_p)$ is cyclic of order n with generator σ_p . [Hint: Remember that

$$|G(E/F)| = [E : F],$$

for a normal field extension E over F .]

Solution to (i). Recall that $\sigma_p : E \rightarrow E$ is defined by $\sigma_p(\alpha) = \alpha^p$. First, we compute $\sigma_p^k(\alpha) = \alpha^{p^k}$ for any $k \geq 1$. Since $|E| = p^n$, every element $\alpha \in E$ satisfies $\alpha^{p^n} = \alpha$, so $\sigma_p^n = \text{id}$. Thus the order of σ_p divides n . Now suppose $\sigma_p^k = \text{id}$ for some $1 \leq k \leq n$. Then $\alpha^{p^k} = \alpha$ for all $\alpha \in E$, meaning every element of E is a root of $x^{p^k} - x$. But $x^{p^k} - x$ has at most p^k roots, while $|E| = p^n$. This forces $p^n \leq p^k$, hence $n \leq k$. Therefore the order of σ_p is exactly n . $\square$

Solution to (ii). Since E is a finite field of order p^n , it is a splitting field of $x^{p^n} - x$ over $\mathbb{Z}_p$, so $E/\mathbb{Z}_p$ is a normal extension with $[E : \mathbb{Z}_p] = n$. By the hint,

$$|G(E/\mathbb{Z}_p)| = [E : \mathbb{Z}_p] = n.$$

By part (i), $\sigma_p \in G(E/\mathbb{Z}_p)$ has order n . Since $G(E/\mathbb{Z}_p)$ is a group of order n containing an element of order n , it is cyclic with generator σ_p :

$$G(E/\mathbb{Z}_p) = \langle \sigma_p \rangle \cong \mathbb{Z}_n. \quad \square$$

Exercise 46.11. Let K be the splitting field of $x^3 - 5$ over $\mathbb{Q}$.

- (i) Show that $K = \mathbb{Q}(\sqrt[3]{5}, i\sqrt{3})$.
- (ii) Describe the six elements of $G(K/\mathbb{Q})$ by giving their values on $\sqrt[3]{5}$ and $i\sqrt{3}$.
- (iii) To what group we have seen before is $G(K/\mathbb{Q})$ isomorphic?
- (iv) Using the notation given in the answer to part (ii) in the back of the text, give the diagrams for the subfields of K and for the subgroups of $G(K/\mathbb{Q})$, indicating corresponding intermediate fields and subgroups, as we did in Example 46.11.

Solution to (i). The roots of $x^3 - 5$ over $\mathbb{Q}$ are $\sqrt[3]{5}$, $\sqrt[3]{5}\omega$, $\sqrt[3]{5}\omega^2$, where $\omega = e^{2\pi i/3}$ is a primitive cube root of unity. Since

$$\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i,$$

we have $i\sqrt{3} = 2\omega + 1$, so $i\sqrt{3} \in \mathbb{Q}(\sqrt[3]{5}, \omega)$. Conversely, $\omega = \frac{-1+i\sqrt{3}}{2} \in \mathbb{Q}(\sqrt[3]{5}, i\sqrt{3})$. Therefore $\mathbb{Q}(\sqrt[3]{5}, \omega) = \mathbb{Q}(\sqrt[3]{5}, i\sqrt{3})$, and since the splitting field of $x^3 - 5$ is $\mathbb{Q}(\sqrt[3]{5}, \omega)$, we conclude $K = \mathbb{Q}(\sqrt[3]{5}, i\sqrt{3})$. $\square$

Solution to (ii). Any $\varphi \in G(K/\mathbb{Q})$ is determined by its action on $\sqrt[3]{5}$ (which must map to one of the three roots $\sqrt[3]{5}\omega^k$) and on $i\sqrt{3}$ (which must map to $\pm i\sqrt{3}$, as it satisfies $x^2 + 3 = 0$ over $\mathbb{Q}$). Note that when $i\sqrt{3} \mapsto -i\sqrt{3}$, we have $\omega \mapsto \bar{\omega} = \omega^2$. This constrains the images of $\sqrt[3]{5}\omega^k$ accordingly. The six elements are:

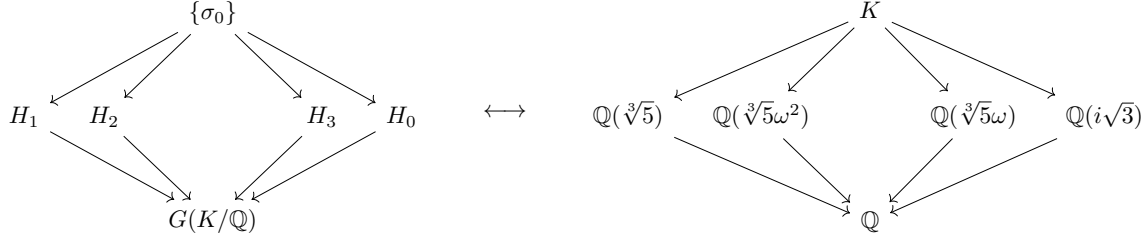
$$\begin{aligned} \sigma_0 : \sqrt[3]{5} &\mapsto \sqrt[3]{5}, & i\sqrt{3} &\mapsto i\sqrt{3}, \\ \sigma_1 : \sqrt[3]{5} &\mapsto \sqrt[3]{5}\omega, & i\sqrt{3} &\mapsto i\sqrt{3}, \\ \sigma_2 : \sqrt[3]{5} &\mapsto \sqrt[3]{5}\omega^2, & i\sqrt{3} &\mapsto i\sqrt{3}, \\ \tau_0 : \sqrt[3]{5} &\mapsto \sqrt[3]{5}, & i\sqrt{3} &\mapsto -i\sqrt{3}, \\ \tau_1 : \sqrt[3]{5} &\mapsto \sqrt[3]{5}\omega^2, & i\sqrt{3} &\mapsto -i\sqrt{3}, \\ \tau_2 : \sqrt[3]{5} &\mapsto \sqrt[3]{5}\omega, & i\sqrt{3} &\mapsto -i\sqrt{3}. \end{aligned} \quad \square$$

Solution to (iii). We know that $|G(K/\mathbb{Q})| = [K : \mathbb{Q}] = 6$, and $G(K/\mathbb{Q})$ is nonabelian since $\sigma_1\tau_0 \neq \tau_0\sigma_1$. Thus, the only group that $G(K/\mathbb{Q})$ can be isomorphic to is the symmetric group S_3 , which is the unique nonabelian group of order 6. Therefore, we conclude that $G(K/\mathbb{Q}) \cong S_3$. $\square$

Solution to (iv). The proper subgroups of $G(K/\mathbb{Q})$ are the unique subgroup of order 3, i.e., $H_0 = \{\sigma_0, \sigma_1, \sigma_2\}$, which is normal in $G(K/\mathbb{Q})$, and the three subgroups of order 2, i.e., $H_1 = \{\sigma_0, \tau_1\}$, $H_2 = \{\sigma_0, \tau_1\}$, and $H_3 = \{\sigma_0, \tau_1\}$.

The fixed field, K_{H_0} is the unique quadratic subfield of K fixed by all σ_k , which is $\mathbb{Q}(i\sqrt{3})$. The fixed field, K_{H_1} is the unique cubic subfield of K fixed by τ_0 , which is $\mathbb{Q}(\sqrt[3]{5})$. The fixed field, K_{H_2} is $\mathbb{Q}(\sqrt[3]{5}\omega^2)$, fixed by τ_1 . The fixed field, K_{H_3} is $\mathbb{Q}(\sqrt[3]{5}\omega)$, fixed by τ_2 .

The diagrams for the subgroups of $G(K/\mathbb{Q})$ and the corresponding subfields of K are:



The three subfields $\mathbb{Q}(\sqrt[3]{5})$, $\mathbb{Q}(\sqrt[3]{5}\omega)$, $\mathbb{Q}(\sqrt[3]{5}\omega^2)$ each have degree 3 over $\mathbb{Q}$ and correspond to the non-normal subgroups H_1, H_2, H_3 of order 2. The subfield $\mathbb{Q}(i\sqrt{3})$ has degree 2 over $\mathbb{Q}$ and corresponds to the normal subgroup H_0 of order 3, consistent with the fact that $\mathbb{Q}(i\sqrt{3})/\mathbb{Q}$ is a normal extension. $\square$

Exercise 46.12. Describe the group of the polynomial $(x^4 - 5x^2 + 6) \in \mathbb{Q}[x]$ over $\mathbb{Q}$.

Solution. Factoring the polynomial, we get $f(x) = x^4 - 5x^2 + 6 = (x^2 - 2)(x^2 - 3)$. Thus, the roots of $f(x)$ are $\pm\sqrt{2}$ and $\pm\sqrt{3}$. The splitting field of $f(x)$ over $\mathbb{Q}$ is $F = \mathbb{Q}(\sqrt{2}, \sqrt{3})$.

Next, we determine $[F : \mathbb{Q}]$. Since $x^2 - 2$ is irreducible over $\mathbb{Q}$, we have $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$. The polynomial $x^2 - 3$ is irreducible over $\mathbb{Q}(\sqrt{2})$ because $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$ (as $\sqrt{3} = a + b\sqrt{2}$ for $a, b \in \mathbb{Q}$ leads to a contradiction). Thus $[F : \mathbb{Q}(\sqrt{2})] = 2$, we have:

$$[F : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})][\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2 \cdot 2 = 4.$$

Since $x^4 - 5x^2 + 6$ is separable and F is its splitting field, the extension $F/\mathbb{Q}$ is Galois, and $|G(F/\mathbb{Q})| = 4$.

The automorphisms in $G(F/\mathbb{Q})$ are determined by their action on $\sqrt{2}$ and $\sqrt{3}$, each of which can be sent to its negative independently. Define:

$$\begin{aligned} \sigma : \sqrt{2} &\mapsto -\sqrt{2}, & \sqrt{3} &\mapsto \sqrt{3}, \\ \tau : \sqrt{2} &\mapsto \sqrt{2}, & \sqrt{3} &\mapsto -\sqrt{3}. \end{aligned}$$

Then $G(F/\mathbb{Q}) = \{\text{Id}, \sigma, \tau, \sigma\tau\}$, where $\sigma^2 = \tau^2 = \text{id}$ and $\sigma\tau = \tau\sigma$. Therefore,

$$G(F/\mathbb{Q}) \cong \mathbb{Z}_2 \times \mathbb{Z}_2. \quad \square$$